

CRANE LAKE SCOTTS SPB, MN, USA

WMO: 727473

Lat: 48.267N

Lon: 92.487W

Elev: 341

StdP: 97.29

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94977

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.9	(c) -29.9	(d) -37.2	(e) 0.1	(f) -33.0	(g) -32.9	(h) 0.2	(i) -29.6	(j) 6.6	(k) -9.7	(l) 5.7	(m) -10.0	(n) 0.1	(o) 160	(p) 0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	29.8	20.7	27.8	19.7	26.9	19.1	23.0	27.5	21.8	26.2	20.5	24.4	2.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.4	25.7	20.2	15.5	24.1	18.9	14.3	22.9	70.1	27.7	65.0	26.1	60.5	24.7	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.0	5.2	4.6	-36.2	32.7	3.7	2.1	-38.9	34.2	-41.0	35.4	-43.1	36.6	-45.8	38.1
			-36.3	25.2	3.6	1.6	-38.9	26.4	-41.0	27.3	-43.0	28.2	-45.6	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-14.1	-12.4	-4.6	3.1	10.3	16.0	18.5	17.3	13.0	5.7	-2.4	-10.6		
	DBStd	12.85	8.22	7.83	7.50	5.38	4.72	3.97	3.27	3.57	4.63	4.84	6.07	7.57		
	HDD10.0	3282	746	628	455	215	54	3	0	0	21	151	372	638		
	HDD18.3	5558	1004	861	712	458	253	91	39	63	168	391	621	896		
	CDD10.0	874	0	0	1	7	63	183	263	226	112	19	1	0		
	CDD18.3	107	0	0	0	0	3	21	43	30	9	1	0	0		
	CDH23.3	1328	0	0	0	4	84	269	508	348	108	7	0	0		
	CDH26.7	326	0	0	0	1	20	69	129	86	21	1	0	0		
(14) Wind	WSAvg	1.5	1.3	1.4	1.5	1.7	1.7	1.3	1.3	1.2	1.5	1.6	1.6	1.3		
(15) Precipitation	PrecAvg	706	22	17	28	50	76	101	102	81	94	65	38	28		
	PrecMax	854	50	34	85	111	130	282	204	157	188	110	87	59		
	PrecMin	588	9	3	10	15	29	44	42	18	30	15	2	14		
	PrecStd	78	11	9	17	22	27	51	47	37	46	28	22	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	7.5	17.2	22.6	29.0	32.2	32.5	32.1	29.0	23.8	14.8	7.1		
		MCWB	1.8	3.3	10.2	12.4	16.9	22.7	22.8	23.0	20.0	15.1	10.3	3.0		
	2%	DB	2.1	4.0	12.2	18.9	26.0	28.0	29.1	28.8	26.4	20.0	11.4	2.9		
		MCWB	0.8	1.6	7.2	9.9	15.1	19.3	20.9	20.7	18.9	13.2	8.0	1.3		
	5%	DB	-0.1	2.2	8.5	16.2	22.6	26.3	27.7	27.2	23.9	17.1	8.0	1.4		
		MCWB	-0.9	0.7	4.6	8.4	13.7	18.0	20.1	19.9	17.7	12.3	5.4	0.5		
	10%	DB	-2.3	-0.2	6.0	12.8	20.1	24.1	26.3	25.1	22.1	13.7	6.2	-0.2		
		MCWB	-3.1	-1.0	2.6	6.6	12.9	17.1	19.4	18.4	16.6	10.1	4.0	-0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	4.3	12.2	14.3	20.0	23.5	24.6	24.7	22.0	17.6	12.1	3.4		
		MCDB	4.1	6.4	14.3	20.5	23.8	28.8	29.0	29.1	26.9	20.3	13.5	5.9		
	2%	WB	0.7	2.1	7.7	11.3	17.4	21.7	23.2	22.8	20.4	14.9	8.0	1.8		
		MCDB	1.7	3.9	11.4	16.9	22.0	26.6	27.3	27.1	24.6	18.9	10.1	2.9		
	5%	WB	-1.1	0.5	4.9	9.2	15.6	20.2	21.9	21.1	19.0	12.6	6.1	0.5		
		MCDB	-0.4	2.0	8.3	15.2	20.8	24.0	26.0	25.2	22.1	16.2	8.1	1.5		
	10%	WB	-2.9	-1.4	2.7	7.3	13.8	18.8	20.6	19.7	17.5	10.8	3.9	-1.0		
		MCDB	-2.3	-0.3	5.0	12.2	18.3	22.4	24.4	23.5	20.4	13.3	6.0	-0.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	13.6	13.2	13.1	13.6	12.9	13.0	13.0	12.4	9.8	8.1	9.4		
		MCDBR	10.6	13.1	15.1	18.6	18.6	15.6	14.9	14.9	15.7	15.3	11.7	10.0		
	5% WB	MCWBR	9.2	10.4	9.9	10.8	9.7	8.1	8.0	8.0	8.9	9.1	8.3	8.2		
		MCWBR	9.6	11.4	13.9	16.5	15.2	12.7	13.3	13.2	12.6	13.2	10.9	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.270	0.321	0.337	0.358	0.375	0.389	0.372	0.365	0.308	0.287	0.262		
	taud		2.347	2.316	2.325	2.419	2.412	2.387	2.371	2.448	2.437	2.584	2.508	2.390		
	Ebn at Noon		827	896	895	915	904	887	867	869	840	851	792	775		
	Edh at Noon		68	89	105	106	112	115	116	103	94	69	60	58		
(44) All-Sky Solar Radiation	RadAvg		1.33	2.51	3.73	4.74	5.25	5.58	5.92	5.01	3.70	2.16	1.20	0.94		
	RadStd		0.17	0.26	0.35	0.44	0.35	0.44	0.32	0.35	0.30	0.30	0.15	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page